

## Technology Stack

|                      |                                             |
|----------------------|---------------------------------------------|
| <b>Date</b>          | 15 March 2026                               |
| <b>Team ID</b>       | XXXXXX                                      |
| <b>Project Name</b>  | Emotion Detection & Learning Support Engine |
| <b>Maximum Marks</b> | 2 Marks                                     |

### Define Your Technology Stack:

The Emotion Detection & Learning Support Engine is developed using a modern technology stack that combines artificial intelligence, natural language processing, and web technologies. The selected tools and frameworks ensure accurate emotion detection, personalized learning support, scalability, maintainability, and a user-friendly experience. The table below lists the technologies used in each layer of the system along with their purpose and justification.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen  | Justification / Purpose                                                                                |
|------|--------------------------------|--------------------|--------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | Streamlit (Python) | Provides a simple, interactive, and responsive web interface for students to interact with the system. |

|   |                                     |                                    |                                                                                                         |
|---|-------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------|
| 2 | <b>Backend / Server-Side</b>        | <i>Python</i>                      | <i>Supports machine learning, NLP processing, and seamless integration with AI models and APIs.</i>     |
| 3 | <b>Database / Data Storage</b>      | <i>CSV Files (Emotion History)</i> | <i>Lightweight storage for user inputs, detected emotions, and learning history during development.</i> |
| 4 | <b>Cloud / Hosting / Deployment</b> | <i>Streamlit Community Cloud</i>   | <i>Enables easy deployment and access through a web browser with minimal configuration.</i>             |
| 5 | <b>Version Control &amp; CI/CD</b>  | <i>Git &amp; GitHub</i>            | <i>Provides version control, collaboration, and secure project management</i>                           |

|          |                                           |                                                                                 |                                                                                                                                                                                                                                   |
|----------|-------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6</b> | <b>Third-Party APIs /<br/>Other Tools</b> | <i>Gemini API,<br/>BiLSTM, BERT,<br/>TensorFlow,<br/>Transformers,<br/>NLTK</i> | <i>Gemini API generates<br/>personalized learning<br/>recommendations, while<br/>BiLSTM and BERT detect<br/>emotions from text.<br/>TensorFlow, Transformers,<br/>and NLTK support model<br/>training and NLP<br/>processing.</i> |
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